

# Web e-learning: Automated Essay Assessment Based on Natural Language Processing Using Vector Space Model

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**Abstract:** Automatic essay assessment aims to determine the value of essay and provides feedback based on a computer system. Using a computer-based system can make essay assessments more objective so that the students are more satisfied with the results they get. The assessment of the learning outcomes is very important for web e-learning systems. The application of the right algorithm can provide the right assessment results. This article aims to develop a web e-Learning system with the automatic essay answer scoring application based on natural language processing (NLP) with the text processing. It consists of tokenizing, stopword removal and stemming by using a vector space model algorithm to calculate the word weights. The feasibility validation test phase of application system was carried out by comparing between manual assessment and automatic assessment. The results of manual calculation shows the similarity with a percentage of 82%. while the level of similarity of the automatic calculation system is 81.6%. The calculation value of similarity to all sample questions tested has a significant proximity

**Keywords:** Automatic Essay Assessment, Natural Language Processing, Vector Space Model, Algorithm

## I. INTRODUCTION

The modernization of technology is getting more progressed and effective at all times. For example in the education, there has been a very significant modernization with the use of E-Learning[1]. The assessment process on the web e-learning is very important to be carried out appropriately. The assessment can be done online with an essay question model[2]. The essay model takes more time to assess it, especially if the number of questions and students taking the exam is large, sometimes the assessment is less objective. The subjective nature of essay assessment causes assessment variety in the class that are carried out to be different. This is considered unfair assessment by the students. Therefore, It needs a method for evaluating essays automatically, which can compare word by word between the answer keys provided by the lecturer and the answers of the students.

### A. Natural Language Processing (NLP)

One part of Artificial Intelligence (AI) is Natural Language Processing which makes computer applications that is able to understand human natural language, such as text and voice that are often used by humans in daily conversation[3][4]. Linguistic structure in NLP is important

in the process of understanding conversation, but it sometimes is overlooked[5]. Therefore, the algorithm used must be able to make the conversation between the user and the natural system. Along with the advancement of the computer interfaces, NLP has been developed in education aspect. This technique makes the computer able to do the tasks of the lecturers such as checking the answers of essays, etc.[6]. Implementing NLP in an educational environment is important for improving the academic performance[7][8]

### B. Vector Space Model (VSM)

Vector Space Model is “a method to see the level of closeness of the similarity terms by means of term weighting[9]. Document is seen as a vector that has magnitude and direction”[10][11]. In the Vector Space Model, a term is represented by a dimension of the vector space. The relevance of a document to a query is based on the similarity between the document vector and the query vector[12][13]. The process of calculating the Vector space model “[14] is started by calculating the term frequency (tf) with the equation:

$$tf = tf_{ij} \quad (1)$$

where  $tf$  = term frequency,  $tf_{ij}$  = the number of occurrences of term  $t_i$  in the  $d_j$  document

Calculate Inverse Document Frequency (idf), with the equation:

$$idf_i = \log \frac{N}{df_i} \quad (2)$$

where  $idf_i$  = inverse document frequency,  $N$  = number of documents retrieved by the system,  $df_i$  = number of documents in collection where the term  $t_i$  appears in it

Calculating term frequency Inverse Document Frequency( $tf \cdot idf$ ), with the equation:

$$W_{ij} = tf_{ij} \times \log \frac{N}{df_i} \quad (3)$$

where  $ij$  = document weight,  $N$  = Number of documents retrieved by the system,  $tf_{ij}$  = the number of occurrences of

the term  $t_i$  in the  $d_j$  document,  $df_i$  = number of documents in collection where the term  $t_i$  appears in it.

Calculating query distance, equation:

$$|q| = \sqrt{\sum_{i=1}^t (W_i \cdot iq)} \quad (4)$$

where  $|q|$ =Query distance,  
 $Wiq$ =document query weight to- $i$

Calculating Document Distance, with the equation:

$$|dj| = \sqrt{\sum_{i=1}^t (Wij)^2} \quad (5)$$

where  $|dj|$  = document distance,  
 $Wij$  = document weight to- $i$

Calculating index terms from documents and queries ( $q, d_j$ ) with the equation 6:

$$q, d_j = \sum_{i=1}^t W_{i,q} \times W_{i,j} \quad (6)$$

where  $W_{ij}$  = weight of term in document,  
 $W_{iq}$  = query weight

### C. Cosine Similarity

The similarity between two or more documents is calculated by using the function of a similarity measurement [15]. "The bigger result of the similarity function, the more similarity of the two objects evaluated, and vice versa. This measuring enables to the rank of the documents according to their similarity (relevance) to the query. The results quality of the documents obtained is dependent on the similarity function used [16].

The cosine value of the angle from the two vectors product is compared because the cosine of 0 is 1 and less than 1 for other angle values, then the similarity value of the two vectors is analyzed similar when the value of the cosine similarity is 1". Here is the formula for the cosine similarity[17]:

$$\text{Similarity} = \cos(\theta) = \frac{Q \cdot D}{|Q| |D|} = \frac{\sum_{i=1}^N (wqi \times wdi)}{\sqrt{\sum_{i=1}^N (wqi)^2} \times \sqrt{\sum_{i=1}^N (wdi)^2}} \quad (7)$$

Where:

$Q \cdot D$  = dot product between vector  $Q$  and vector  $D$ ,  
 $|Q|$  = length of vector  $Q$ ,  
 $|D|$  = length of vector  $D$ ,  
 $|Q||D|$  = cross product between  $|Q|$  and  $|D|$ ,  
 $wqi$  = term weight on query to  $I$ ,  
 $w dij$  = weight of term in document  $i$  term,  
 $N$  = number of vectors

### D. Assessment

Assessment is an effort to obtain data/information from the learning process and results to find out how well the student's performance are compared to certain learning outcomes[18]. Assessing students and providing timely feedback are important aspect of learning[19]. After obtaining the of the assessment results, the grading process

is done. Grading is the process of embedding attributes or dimensions or quantities (in the form of numbers/letters) to the results of the assessment by comparing them to a certain standard instrument. The assessment results in the form of attributes/dimensions/ quantities are used as evaluation materials[20].

## II. METHOD

This research uses the following activities: 1) problem analysis; 2) needs analysis; 3) development web e-learning utomated Essay Assessment. The results of the development were validated by the experts. The trial was conducted with exercise data (sample questions). The development process is shown in Fig.1

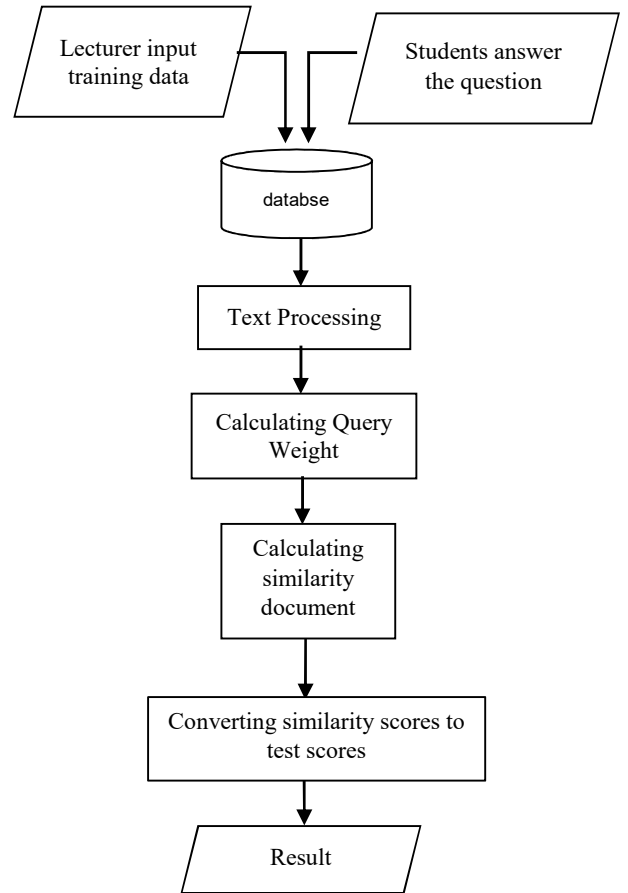


Fig.1 Development process

## III. RESULTS

### A. The results of web e-learning development

The learning content of the web e-learning interface is shown in Figure-2. The lecturers manage the learning content through the interface as shown in Fig. 2. The students choose the content of the material to be learned. Furthermore, the interface for doing essay evaluation is shown in Fig. 3 and the results of the assessment are shown in Fig. 4:

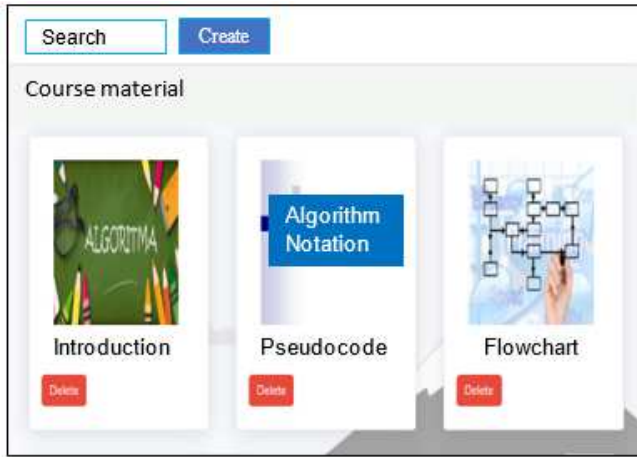


Fig. 2 Learning content interface

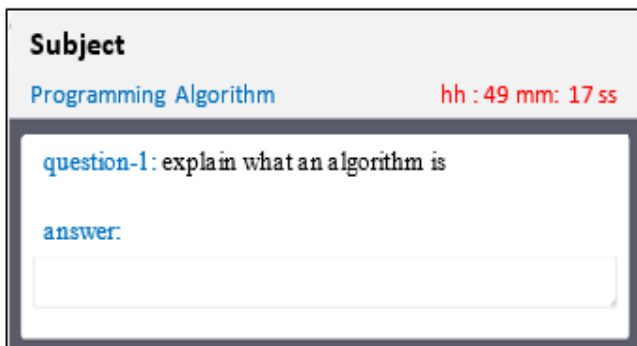


Fig. 3 assessment interface

| Result                     |             |         | Home            |
|----------------------------|-------------|---------|-----------------|
| Show                       | 10          | entries | Search          |
| No                         | Similarites | Score   |                 |
| 1                          | 1           | 20      |                 |
| 2                          | 1           | 20      |                 |
| 3                          | 1           | 20      |                 |
| 4                          | 1           | 20      |                 |
| 5                          | 1           | 20      |                 |
| Sowing 1 to 5 of 5 entries |             |         | Previous 1 Next |

Fig. 4 Assessment results

### B. Similarity Testing

The document similarity testing is a test carried out to measure the document similarity of the manual calculation and automatic calculation results. The following is an example of a manual calculation of the question number-1 and the answers of students.

TABLE I. MANUAL CALCULATION, QUESTION NUMBER 1

| Term              | TF |   | TF' |    | Q^2 | D^2 | Q*D |
|-------------------|----|---|-----|----|-----|-----|-----|
|                   | Q  | D | Q'  | D' |     |     |     |
| algorithm         | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| is                | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| sequence          | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| or                | 1  | 0 | 1   | 0  | 1   | 0   | 0   |
| logical           | 1  | 0 | 1   | 0  | 1   | 0   | 0   |
| steps             | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| solving           | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| computer          | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| programming       | 1  | 0 | 1   | 0  | 1   | 0   | 0   |
| problems          | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| problems          | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| systematically    | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| arranged          | 1  | 1 | 1   | 1  | 1   | 1   | 1   |
| Score = 0,88      |    |   |     |    | 13  | 10  | 10  |
| Conversion = 17.6 |    |   |     |    |     |     |     |

Manual calculations are shown in the form of table 1, the value of Q indicates the answer key, while the value of D indicates the answer from the student. The results of the calculation were found that the value of Q was 13 words while the value of D was 10 words, and then from the results of the multiplication of  $Q \times D$ , the number of words counted was 10 words, then it found the root value of each of the total number of Q and D. The results of the next root were used as the numerator by calculating the total number of words  $Q \times D$  which was divided by the root of Q and the root of D. Then it found the result of a similarity is (0.88) from the total similarity (1.00). The result is then converted by multiplying (value per question), for example for the value per question = 20, then the calculation is  $(20 \times 0.88)$  with a conversion result 17.6.

Based on the 5 samples that have been calculated manually, it can be concluded by using table 2:

TABLE II. MANUAL CALCULATION CONCLUSION

| No.                | Similarity | Percentage | Results |
|--------------------|------------|------------|---------|
| 1                  | 0,88       | 88%        | 17.6    |
| 2                  | 0,82       | 82%        | 16.4    |
| 3                  | 0,77       | 77%        | 15.4    |
| 4                  | 1,00       | 100%       | 20      |
| 5                  | 0,63       | 63%        | 12.6    |
| Average Percentage |            | 82%        |         |

Based on the table above, a comparison is made by using automatic calculations. As in table-3.

TABLE III. AUTOMATIC CALCULATION CONCLUSION

| No.                | Similarity | Percentage | Results |
|--------------------|------------|------------|---------|
| 1                  | 0,88       | 88%        | 17.6    |
| 2                  | 0,9        | 90%        | 18      |
| 3                  | 0,8        | 80%        | 16      |
| 4                  | 1,0        | 100%       | 20      |
| 5                  | 0,5        | 50%        | 10      |
| Average Percentage |            | 81.6%      |         |

The conclusion of 5 samples, there are 3 calculation results that are in accordance with manual calculations, while 2 samples are close to the calculation.

### C. Text Processing Test

Answer: "An algorithm is a sequence or logical steps for solving the computer programming problems that are systematically arranged"

#### a. Tokenizing>Stopward removal

In the tokenizing>stopward removal stage above, it indicates that there are 5 words removed, this means that stopward removal cannot be used or applied to the system.

#### b. Tokenizing>Stemming

By using the tokenizing>stemming process, it can be seen that text processing by using stemming is inaccurate and will make the system in the application interpret 2 different vocabularies to be the same meaning, for example in the sentence above it produces: "An algorithm is a sequence or logical steps to solve a computer program problem that is systematically arranged".

#### c. Tokenizing>Tokenizing

In the text processing that only uses tokenizing as above, no words are omitted so that the accuracy in the word processing is not reduced, and is very supportive in the use of this application.

## IV. CONCLUSION

The results of manual calculations showed that the similarity order of each sample question with a value of 0.88, 0.82, 0.77, 1.00, 0.63 with a percentage of 82%. Comparing with the level of similarity obtained from the automatic calculation system, it was found a value of 0.88, 0.90, 0.80, 1.00, 0.50 with a percentage of 81.6%. Thus, the conclusion of the 5 question samples, there are 3 calculation results that are in accordance with manual calculations, while 2 samples are close to the automatic calculation. The calculation value of similarity to all sample question tested has a significant proximity.

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